

Definition:

The derivative of a function f at a point x_0 (denoted by $f'(x_0)$) is the limit of the ratio of ~~the~~ ^{an} increment of the function f at the point x_0 (denoted by $\Delta f(x_0)$) to an increment Δx of the independent variable for $\Delta x \rightarrow 0$:

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{\Delta f(x_0)}{\Delta x}$$

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \quad (2).$$

Note that you are already familiar with the right-hand side of equation (2) (cf. expression (3) from the previous dialogue).

Reader: Actually the derivative of the function f at point x_0 is the limit of the function $\frac{F}{G}$ at $\Delta x = 0$. The independent variable of the function $\left(\frac{F}{G}\right)$

$$\frac{F}{G} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

is the increment Δx .

Author: You are quite right. However, in what follows you must use the defn. of the derivative as formulated above. This definition does not involve the function $\left(\frac{F}{G}\right)$ of Δx since this function plays, as you understand, only an auxiliary role. We should simply bear in mind that the phrase "the limit of the ratio of an increment $\Delta f(x_0)$ to an increment Δx for $\Delta x \rightarrow 0$ " describes the limit of a function Δx , i.e. the function $\left(\frac{F}{G}\right)$, which is considered at $\Delta x = 0$. The derivative can also be interpreted [in terms of geometry].

Reader: Shall we do it by using again the Tangent to the graph of a function?

Author: Yes, of course. Let us take the graph $y = f(x)$ (Fig 38), fixing a point $x = x_0$. Consider an increment Δx_1 of the argument; the corresponding increment of the function at point x_0 is $\Delta f_1(x_0)$. Denote the slope of the chord AB_1 by α_1 ; it is readily apparent that

$$\frac{\Delta f_1(x_0)}{\Delta x_1} = \tan \alpha_1$$